

Societal Impact of Research and Technology

Three lectures, many exercises, and Project 3

Osama Alshaykh

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START HERE

How inclusion shapes design

Kat Holmes (Microsoft, Google, now Salesforce; author of "Mismatch")



► [Open on YouTube](#)

WATCH FOR

What is a mismatch?

Is it in the person, or in the design?

Who creates it?

And who can remove it?

Why is it good business?

Not charity, not compliance.

Solve for one, extend to many

Kat Holmes' argument: disability is not a property of the person. It is a mismatch between a person and a design. Designers create these mismatches, and designers can remove them.

Designing for the person at the edge makes the product better for everyone in the middle.

PERMANENT

One arm

Designed for the few people who have one arm...

TEMPORARY

Arm injury

...also helps everyone with a broken wrist for six weeks...

SITUATIONAL

Holding a baby

...and every parent holding a baby, a coffee, or a bag.

The same idea applies to seeing (blind, cataract, bright sun), hearing (deaf, ear infection, loud train), and speaking.

► **Watch:** [Be My Eyes: a product built for blind users, now sold to companies as customer service](#)

Built for accessibility, used by everyone

CURB CUTS

Built for wheelchairs. Now used by strollers, suitcases, bikes, and delivery carts.

CAPTIONS

Built for deaf viewers. Now used in noisy places, with the sound off, and by language learners.

AUDIOBOOKS

Started as talking books for blind readers in the 1930s. Now a mainstream industry.

OXO GOOD GRIPS

Kitchen tools designed for a person with arthritis. They became a bestseller because they are easier for everyone.

VOICE CONTROL AND TEXT-TO-SPEECH

Built for people who cannot type or read a screen. Now in every phone and car.

LARGE TEXT AND HIGH CONTRAST

Built for low vision. Now how many of us read our phones outside in the sun.

Discussion: which accessibility features on your phone do you already use?

WHAT WE WILL DO FOR THREE LECTURES

Every design is a decision.

Every mismatch is a decision someone made, usually without noticing: the test set we did not diversify, the label we did not question, the failure we did not log. And every mismatch is a chance to make the product better.

We will look for these decisions in real products. Then you will look for them in yours.

It will be interactive, with many exercises.

Six questions we will ask every time

Every case is tagged with the questions it failed. You will answer all six for your project.

Q1

Who was in the test set?

And who will actually use it? Harm lives in the gap.

Q2

What are we really optimizing?

e.g., cost instead of health need, engagement instead of value.

Q3

What happens at scale?

A 1-in-10,000 failure with 10 million users is 1,000 people.

Q4

Who notices when it fails?

Is it logged and reported? Can anyone check what happened?

Q5

Who can override or appeal?

Does the person in the loop ever disagree with the machine?

Q6

Where does the data go?

Including after an acquisition, a breach, or a bankruptcy.

What I want you to be able to do

1

Trace a mismatch to a design decision

e.g., a sensor, a dataset, a label, a threshold, a default. "The AI is biased" is not an explanation.

2

Judge the evidence

Who paid for the study? Compared to what? Measured how? Has anyone checked it?

3

Measure it in your own product

Which metric, for which group, measured how. No numbers, no claim.

4

Know your obligations

Privacy, AI, and export rules. Where are you selling, and to whom?

5

Improve your product, and speak up

Find one change that makes it work for more people. Know what to do when you see a problem.

The next three lectures follow the life of a product

LECTURE 1

Who is it for?

Accessibility in your app, sensors, a national ID system, and how to read an impact claim

Exercises 1 and 2

LECTURE 2

What does it decide?

Labels, fairness with real data, people trusting machines, from the lab to the field, self-driving cars, scale

Exercises 3 and 4

LECTURE 3

Why does it keep happening?

Incentives, misuse, jobs, research choices, Boeing and you, data, the law, and a method

Exercise 5 and Project 3

LECTURE 1

Who is it for?

Mismatches start with who we design for, and who we test with.

20 MIN

Exercise 1: Make your app better for everyone

Apply Kat Holmes to your own product. Start with your phone's accessibility features, then your team's app, on one core user story.

1

Your phone

List the accessibility features. Which ones do you already use without a disability?

2

Screen reader

Turn on VoiceOver (iOS) or TalkBack (Android). Finish the story without looking.

3

Real situations

One hand only. Bright sunlight. Loud room. Text at 200%. What breaks?

4

One fix

Pick a fix that helps everyone, not only people with a disability. Put it in your backlog.

Report back: what broke, and the fix that makes the product better for all users.

► **Watch:** [Accessibility in Android: users with motor impairments](#)

► **Watch:** [Accessibility in Android: users with visual impairments](#)

Oximeters: the standard was the problem

2020 → 2025

Fails Q1

~3×

more hidden low oxygen missed in Black patients than White patients (NEJM, 2020)

2 of 10

darkly pigmented subjects: the old FDA minimum for a validation study (or 15%)

150+

participants across skin tones in the new FDA draft guidance (January 2025)

THE LESSON

Melanin absorbs the red light the device relies on. The devices passed the standard, but the standard let two people represent everyone with dark skin. Meeting a standard does not mean the product works.

DISCUSSION

A large 2025 study (EquiOx) found oximeters mostly UNDER-estimate oxygen, but over-estimate more often in darker skin. Why do studies disagree? What would you measure?

► **Watch:** [Warm-up \(30 seconds\): the soap dispenser, same physics](#)

► **Watch:** [Annals Animated: racial disparity in pulse oximetry](#)

↗ **Read:** [EquiOx study, American College of Cardiology, March 2025](#)

A master's thesis that changed products

2017 → 2020

Fails Q1

34.7%

error rate for darker-skinned women in commercial gender classifiers, vs. 0.8% for lighter-skinned men

1,270

faces in the test set she built, balanced by skin type and gender, because existing benchmarks were not

2020

IBM leaves general face recognition; Amazon and Microsoft stop selling it to police

THE LESSON

Gender Shades: Joy Buolamwini, an MIT graduate student. She built a balanced test set, measured error per group, and published. Within a year the audited companies had cut the gap. You do not need permission to measure.

DISCUSSION

What would a balanced test set for your project look like? Which group would you measure first, and who would you tell if the gap is large?

► **Watch:** [Joy Buolamwini \(TED\): how I'm fighting bias in algorithms](#)

↗ **Read:** [Gender Shades project site: results and methods](#)

↗ **Read:** [Buolamwini and Gebru, "Gender Shades" paper \(2018\)](#)

Aadhaar: the same mismatch, at national scale

India

Fails Q1 · Q5

WHAT HAPPENED

- India delivers subsidized food to about 800 million people. At the ration shop, your fingerprint must match your Aadhaar biometric ID.
- The goal was to stop theft of rations. In Rajasthan in 2016–17, authentication failed for 20–30% of beneficiaries in most districts.
- Fingerprints worn smooth by age or manual labor often fail. The Wire reported an 88-year-old farmer sent home without rations, and an 83-year-old woman denied rations and pension for 16 months before she died.
- Fallbacks such as one-time passwords exist on paper, but often not at the shop.

THE LESSON

Same physics as the oximeter, but here a failed match means no food. When authentication fails, the fallback is the product.

DISCUSSION

Who is right: officials fighting theft, or people turned away? Design a fallback that serves both.

↗ **Read:** [The Wire: poor and elderly bear the brunt of authentication errors](#)

How to read an impact claim

Five checks we will use for the rest of the course.

1

Who paid for it?

The Veritasium self-driving video was sponsored by BMW. Waymo's own crash data vs. an insurance institute's (IIHS).

2

Compared to what?

Robotaxis vs. human drivers, not vs. perfect drivers. A new model vs. the process it replaces.

3

What is the denominator?

Per mile, per trip, per person? "1 in 10 million flight hours" sounds rare until you multiply by a fleet.

4

How was it measured?

Oximeters: hospital records (2020) and a study that measured skin tone directly (2025) disagree.

5

Has anyone checked it?

A court finding vs. an allegation in a lawsuit. A working paper vs. its critics.

10 MIN

Exercise 2: Where else do we use infrared?

Infrared and optical sensing are everywhere. Each team picks one system and finds its version of the oximeter problem.

1

Pick a system

e.g., smartwatch heart rate,
face unlock,
driver-monitoring camera,
phone camera exposure,
pedestrian detection.

2

Name the failure

What could make it fail, and
for whom? Skin tone, age,
worn fingerprints, glasses,
lighting.

3

Design the test

Who is in the test set? How
many people? What metric?
What is a pass?

4

Find evidence

Is there a published study or
a recall? Apply the five
checks to it.

Report back: one sentence per step.

LECTURE 2

What does it decide?

From labels and fairness to what happens when the system leaves the lab.

Algorithms in healthcare

2019 / 2023

Fails Q2

WHAT HAPPENED

- A widely used hospital algorithm predicted health-care cost and used it as a stand-in for health need (Obermeyer et al., Science, 2019).
- Less money is spent on Black patients with the same illness. So at the same score, Black patients were sicker.
- When they predicted need instead, the share of Black patients flagged for extra care went from 17.7% to 46.5%.
- A 2023 lawsuit alleges UnitedHealth used an algorithm (nH Predict) to cut elderly patients' care. Not decided yet.

THE LESSON

The model was accurate. It accurately predicted the wrong thing. Picking the label is an ethical decision.

DISCUSSION

What is the label in your project? What does it actually stand in for?

► **Watch:** [PBS NewsHour: biased algorithms in hospitals](#)

Justice: COMPAS and the impossibility result

2016

Fails Q2 · Q5

WHAT HAPPENED

- ProPublica: among people who did NOT reoffend, Black defendants were about twice as likely to be labeled high-risk (45% vs. 23%).
- The vendor: the scores are calibrated. A score of 7 means the same risk for both groups.
- Both are true. It is proven (Kleinberg 2016, Chouldechova 2017) that when base rates differ, you cannot equalize both calibration and error rates.
- So "make it fair" is not enough. Someone must choose which fairness, and who carries the errors.

THE LESSON

Fairness tools cannot pick the definition for you. It is a values decision. Write it down, and have someone own it.

DISCUSSION

You are the defendant. Which definition do you want? Now you are the victim of the next crime. Did you change your answer?

► **Watch:** [Hany Farid \(TEDx\): algorithms in the courtroom](#)

► **Watch:** [CNBC: how AI could reinforce biases in criminal justice](#)

25 MIN

Exercise 3: Find the impossibility in real data

ProPublica published the COMPAS data for Broward County, Florida. Load compas-scores-two-years.csv. You may use an LLM to write the code; the thinking is yours.

1

Load

Keep African-American and Caucasian. ProPublica's filters:
days_b_screening_arrest
from -30 to 30, is_recid !=
-1, c_charge_degree != "O".
High risk: decile_score ≥ 5 .

2

Error rates

Among people who did NOT
reoffend, what share were
labeled high risk, in each
group?

3

Calibration

For each score from 1 to 10,
what share reoffended, in
each group? Plot both lines.

4

Decide

Both results are true. Which
would you fix, and at what
cost? Write it for a judge.

Report back: your two numbers, your plot, and your recommendation.

↗ **Read:** [ProPublica data on GitHub](#)

↗ **Read:** [ProPublica: how we analyzed the COMPAS algorithm](#)

Fairness tools: what they can and cannot do

THEY CAN

- Compute error rates and selection rates for each group
- Show which fairness definitions your model violates, and by how much
- Compare ways to reduce the gap
- Make the tradeoff visible to a manager

THEY CANNOT

- Choose which fairness is right (Exercise 3)
- Fix a bad label (the healthcare algorithm)
- Check groups you never recorded
- Tell you whether the system should exist

↗ **Read:** [Aequitas, open-source bias audit toolkit \(thanks to Ronnakorn Rattanakornphan for pointing it to me\)](#)

↗ **Read:** [IBM AI Fairness 360](#)

▶ **Watch:** [Google: machine learning and human bias](#)

▶ **Watch:** [MIT 6.S191: algorithmic bias and fairness \(Ava Soleimany\)](#)

People trust the machine

2020 → 2026

Fails Q4 · Q5

WHAT HAPPENED

- Detroit: Robert Williams (2020) and Porcha Woodruff (2023, eight months pregnant) were arrested after false face-recognition matches. The officer told Williams "the computer" said it was him.
- Since the June 2024 settlement, Detroit police cannot arrest on face recognition alone.
- Hiring: in Mobley v. Workday, an applicant over 40 was rejected by 100+ employers using Workday's screening, often within minutes. The court said the vendor could be liable as the employers' "agent". No ruling on the merits yet.

THE LESSON

"A human reviews every decision" is not a safeguard if the human trusts the machine. And "we only sell the tool" may not protect the vendor.

DISCUSSION

Design the screen an officer or a recruiter sees. What would make them doubt the machine?

► **Watch:** [Democracy Now: Porcha Woodruff, arrested while eight months pregnant \(2023\)](#)

↗ **Read:** [Akin: court allows claims against AI hiring tool](#)

When governments automate

1999 → 2025

Fails Q4 · Q5

UK POST OFFICE: HORIZON

Accounting software with known bugs showed cash missing at post office branches. From 1999 to 2015, more than 900 branch operators were prosecuted. The courts assumed the computer was right. Convictions are still being overturned.

NETHERLANDS: CHILDCARE BENEFITS

A fraud risk model at the tax authority flagged families, and dual nationality was one of the risk factors. Tens of thousands of parents were wrongly told to pay back benefits. The Dutch government resigned over it in January 2021.

THE LESSON

Neither system was advanced AI. Nobody could check what the software did, and the people accused had no way to challenge it. Keeping evidence and building appeal are features.

► **Watch:** [ITV News \(free\): what actually happened in the Post Office Horizon scandal](#)

↗ **Read:** [Communications of the ACM: how software bugs led to the scandal](#)

From the lab to the clinic

2018 → 2022

Fails Q1 · Q3

WHAT HAPPENED

- Google's model detected diabetic eye disease with over 90% accuracy in the lab. In Thailand, patients can wait weeks for a specialist to read their eye photos.
- Field study in 11 Thai clinics (2018–19): about 21% of images were rejected as ungradable, mostly because of poor lighting. Slow internet delayed uploads.
- Google published what went wrong (2020) and redesigned the workflow with the nurses.
- A later study in Thailand's national screening program: 7,651 patients, 94.7% accuracy for vision-threatening disease, with specialists checking every image.

THE LESSON

The model was not the product. The product was the model, the camera, the lighting, the network, and the nurse's day. Testing in the field, and publishing the failure, is how it got better.

DISCUSSION

Where will your product actually be used? What is the "clinic lighting" of your project?

↗ **Read:** [Google: healthcare AI systems that put people at the center \(2020\)](#)

↗ **Read:** [Real-time screening in Thailand's national program \(2022\)](#)

Self-driving cars: beyond the trolley problem

2023 → 2026

Fails Q3 · Q4

WHAT HAPPENED

- Cruise, October 2023: a human driver knocked a pedestrian into a robotaxi's path. The robotaxi then dragged her about 20 feet while pulling over.
- Cruise did not show regulators the full video at first. California suspended its permit, and GM shut down the robotaxi business in December 2024.
- Waymo, December 2025: a software recall after its cars passed stopped school buses. The Austin school district documented 19 cases.
- 2026: an insurance institute study (IIHS) found Waymo crashed far less than human drivers in four cities. A CNN review of public records found hundreds of close calls. Both are true.

► **Watch:** [IIHS study \(Aug 2026\): Waymo crashes 68% less than people](#)

► **Watch:** [CNN \(June 2026\): Waymo incidents raise new safety concerns](#)

► **Watch:** [Classic: TED-Ed, the ethical dilemma of self-driving cars](#)

THE LESSON

The real failures are edge cases and how companies report them. The trolley problem makes a nice video, but it is not where cars fail.

DISCUSSION

Is a car that is safer on average, but fails in new and strange ways, acceptable? Apply the five checks.

15 MIN

Exercise 4: Self-driving cars in 20 years?

First question: do you believe it will happen? Then assume half of all rides in a city are driverless. What is the impact on...

1

The car industry

Do we need as many cars per home? Who builds them?

2

Uber, taxis, rentals

What happens to the drivers? To car rental companies?

3

Insurance and liability

Who is liable in a crash: the owner, the maker, or the software company?

4

What else?

City parking and ticket revenue, public transit, people who cannot drive, organ donation.

Report back: one positive and one negative effect you did not think of before.

At scale, rare becomes certain

2024

Fails Q3

8.5 million

Windows computers crashed on July 19, 2024 by one bad CrowdStrike security update. Airlines stopped, hospitals delayed surgeries, some 911 centers went down.

► **Watch:** [CBS News: what caused the outage](#)

THE LESSON

One update, pushed to everyone at once, with deep access to the operating system. A staged rollout to 1% first would have turned a global outage into a small one.

DISCUSSION

If your product fails once in 10,000 uses, how many people is that in its first year? Who are they?

LECTURE 3

Why does it keep happening?

Incentives, research choices, and what you can do about it.

Incentives: what are we optimizing?

2026

Fails Q2

WHAT HAPPENED

- March 2026: a Los Angeles jury found Meta and YouTube negligent for designing features that hooked a young user, e.g., infinite scroll, autoplay, notifications. Both companies say they will appeal.
- The day before, a New Mexico jury ordered Meta to pay \$375 million for misleading users about the safety of its platforms.
- Scientists still debate how much social media harms teen mental health. The juries judged design and warnings, not that debate.

THE LESSON

Engagement is a proxy, just like cost was for health. When the business optimizes the proxy, the product can work perfectly and still harm its users.

DISCUSSION

What metric does your team optimize? What does a user harmed by that metric look like?

► **Watch:** [Global News: Meta and YouTube found liable](#)

↗ **Read:** [Al Jazeera: what we know about the verdict](#)

Generative AI: the users did the testing

2024 → 2026

Fails Q1 · Q3

WHAT HAPPENED

- December 2025 to January 2026: Grok on X let users "undress" photos of real people, including children. One watchdog estimated about 23,000 sexualized images of children in 11 days.
- X added restrictions only after the public outcry. The UK regulator (Ofcom) and the European Commission opened investigations in January 2026.
- Hong Kong, 2024: an employee at the engineering firm Arup sent about US\$25 million after a video call where everyone else was a deepfake.

THE LESSON

Shipping fast is an incentive too. Testing for misuse before launch is cheap. After launch, the public does the testing, and the victims never used the product.

DISCUSSION

Write the misuse test plan you would require before shipping an image-editing feature. Who is on the testing team?

↗ **Read:** [The Register: Ofcom investigation of Grok](#)

Jobs: who captures the value?

2025 → 2026

−19%

is where employment of 22 to 25 year-olds in the most AI-exposed jobs (e.g., software, customer service) stands, compared with their peers in less exposed jobs. The gap keeps widening.

Brynjolfsson, Chandar and Chen, Stanford, revised August 2026, ADP payroll data. It was 13% in August 2025. They also find no economy-wide job loss.

↗ **Read:** [Stanford, August 2026 update](#)

↗ **Read:** [Counterpoint: Economic Innovation Group, January 2026](#)

WHAT ELSE THEY FOUND

Older workers in the same jobs were stable or growing. Companies are hiring fewer juniors rather than paying them less. The drop is where AI replaces tasks, not where it helps people do them.

DISCUSSION

Apply the five checks: other economists disagree. If companies stop hiring juniors, where will the senior engineers of 2035 come from? Whose first job is your product automating?

Research: decisions before you publish

2019 → 2023

Fails Q1 · Q6

WHAT HAPPENED

- 2019: OpenAI released its GPT-2 language model in stages, citing misuse risk. Critics called it hype. Either way, it treated release as a decision.
- 2019: Microsoft took down MS-Celeb-1M, a face dataset of about 10 million photos of 100,000 people, collected without their consent.
- 2022: researchers flipped a drug-discovery model to reward toxicity. It proposed 40,000 potentially lethal molecules in under six hours.
- 2023: LAION-5B, a dataset used to train image generators, was taken offline after Stanford researchers found child sexual abuse material in it.

THE LESSON

Your dataset, your model, and your paper are products too. Ask the six questions before you publish, not after.

DISCUSSION

Your EC601 project uses scraped or public data. Whose consent do you have? What is the worst use of what you release?

Boeing 737 MAX: when engineers see the problem

2018 → 2025

Fails Q3 · Q4 · Q5

WHAT HAPPENED

- Bigger engines changed how the 737 MAX handled. Boeing added software (MCAS) that pushed the nose down automatically.
- MCAS read only one of the plane's two angle-of-attack sensors. Boeing's safety analysis did not rate an MCAS failure as catastrophic, so one sensor was allowed, and it assumed pilots would react within seconds. Pilots were not told about MCAS in their manuals.
- Lion Air (2018) and Ethiopian Airlines (2019) crashed: 346 people died.
- November 2025: a judge dismissed the criminal case under a \$1.1 billion deal, while criticizing it for not holding Boeing accountable.

THE LESSON

A single point of failure is a design decision. So is staying quiet about it. Write concerns down with data, raise them in writing, and escalate. The IEEE Code of Ethics puts public safety first.

DISCUSSION

Two weeks before launch you find a single point of failure. Your manager says ship. What do you do, in order?

↗ **Read:** [Engineering.com: how MCAS came to rely on one sensor](#)

↗ **Read:** [Al Jazeera: judge approves dropping the criminal case \(Nov 2025\)](#)

Where does the data go?

2023 → 2025

Fails Q6

23andMe: the genetic data of about 15 million people, followed through its life.

COLLECT

Saliva and DNA. Your DNA also describes relatives who never agreed.

STORE

2023: a breach exposed data on about 7 million customers.

SHARE

About 80% of customers agreed to research use (company figure).

SELL

March 2025: bankruptcy. June 2025: sold for \$305 million to a nonprofit led by the co-founder.

DELETE

The right to delete was kept, after dozens of states objected.

THE LESSON

Companies like this are generally not covered by HIPAA, the US health privacy law. A privacy policy is only as good as the next owner. Collect less, and design for deletion from day one.

↗ **Read:** [MobiHealthNews: judge approves the 23andMe sale](#)

Encryption: from Apple v. FBI to the UK

2016 → 2026

Fails Q6

WHAT HAPPENED

- 2016: the FBI asked a court to force Apple to unlock a San Bernardino attacker's iPhone. It dropped the case after a third party unlocked the phone.
- 2025: the UK government secretly ordered Apple to provide access to encrypted iCloud data. Apple turned off its Advanced Data Protection feature for UK users instead.
- After US objections, the UK dropped the worldwide demand and reportedly issued a UK-only one. Apple challenged it in court in July 2026.

► **Watch:** [60 Minutes: Tim Cook and James Comey on the iPhone case](#)

↗ **Read:** [CyberInsider: Apple challenges the UK order \(2026\)](#)

THE LESSON

You cannot build a back door that only the good guys can use. Once it exists, it exists for every attacker who finds it.

DISCUSSION

You run a messaging app used by journalists and by criminals. A government asks for access. What do you build?

Classify your product

The law is the floor, not the goal. Most of your projects fall under at least one of these.

Area	Ask	Key rules
Privacy	Personal, health, student, or children’s data? Which users, in which countries and languages?	GDPR (EU General Data Protection Regulation), CCPA/CPRA (California), HIPAA (health), FERPA (student records), COPPA (children under 13)
AI	Does it make decisions about people, e.g., hiring, credit, education?	EU AI Act: bans since Feb 2025; general-purpose AI rules Aug 2025; high-risk systems Dec 2, 2027 (delayed in July 2026)
Security and export	Does it use encryption, quantum, or advanced computing?	US Export Administration Regulations (EAR): encryption controls; quantum computing controls (Sept 2024)

↗ **Read:** [DLA Piper: data protection laws of the world](#)

↗ **Read:** [IAPP global privacy directory](#)

↗ **Read:** [BIS encryption controls](#)

↗ **Read:** [BIS quantum computing controls \(2024\)](#)

15 MIN

Exercise 5: Classify your own product

Fill in the table for your team's project. Be specific: name the data field, the user, the country, and the language. Finish it for Project 3.

1

Data

Every piece of personal data you collect. Where is it stored, and for how long?

2

Users and markets

Children? Patients?
Students? Which countries and languages now, and in five years?

3

Rules

For each area: applies, does not apply, or not sure. "Not sure" needs a specific question.

4

One change

What will you change this sprint because of this exercise?

Report back: the table and your one change. This becomes part of Project 3.

A method in five stages

Each stage, and the cases in these lectures that skipped it.

1

ANTICIPATE

The six questions. A pre-mortem: imagine it failed, then ask why. Brainstorm misuse.

Skipped: Grok

2

DOCUMENT

Write down the label and why. Datasheets for datasets. Model cards.

Skipped: the hospital cost label

3

TEST

Per-group error rates. Accessibility. Field pilots where it will actually be used.

Skipped: oximeters, Aadhaar

4

RELEASE

Staged rollout, starting small. Monitoring. A way to switch it off.

Skipped: CrowdStrike

5

RESPOND

Incident reports. Honest disclosure. A real path to appeal.

Skipped: Cruise, the Post Office

AT EVERY STAGE

Keep the evidence. Could someone outside your team check what your system did, and why?

PROJECT 3

Societal impact of your project

Work on your own team's project. If it really does not affect people, pick one of the alternatives.

Project 3: what you submit

A detailed paper, 8 to 12 pages, with evidence. Due November 24.

1 Who is affected

Users, people affected who are not users, and whoever pays when it fails. Where are the mismatches?

2 The six questions

Answer all six for your product, with specifics.

3 One measurement

Your Exercise 1 accessibility test, error rates per group, or a map of your data. Evidence, not opinion.

4 Classification

Your Exercise 5 table, with sources.

5 Two improvements

Proposed, with an owner, placed in the five-stage method. At least one implemented and shown.

6 Adversarial review

Two different LLMs attack your paper. Submit the logs and your answer to each objection.

↗ **Read:** [Selection of last year's reports](#)

Alternatives

Each one still needs a measurement you did yourself.

OPTION 1: ACCESSIBILITY

Test a leading app with a screen reader and in real situations. Compare with its app-store reviews. Where did accessibility make it better for everyone?

OPTION 2: DATA BIAS

Run a per-group evaluation on a public dataset with Aequitas, AI Fairness 360, or Fairlearn. Show the fairness tradeoff in your own numbers.

OPTION 3: MACHINE LEARNING IN JUSTICE AND POLICING

Analyze one deployed system, e.g., face recognition, risk scores, license-plate readers. Include what happened in court.

OPTION 4: PRIVACY AND SECURITY

Map the full data life of a health, genetic, or IoT (Internet of Things) product, including what happens if the company is sold or goes bankrupt.

What "going deep" means

Everyone who does this gets an A. Keep revising until it is true.

	Not deep	Deep
Evidence	"AI can be biased."	"Our model misses 14% of users over 60 and 5% of others, on 400 test cases."
Specific	"Some users may be excluded."	"TalkBack users cannot check out. The Pay button has no label."
Other side	Only lists harms.	The best argument for shipping anyway, including who is hurt if we do not ship.
Action	"Companies should be careful."	"We labeled the button. The retest passes. Here is the commit."

Videos for these lectures

RECENT (2023 TO 2026)

- ▶ **Watch:** [IIHS: Waymo crashes 68% less than people \(2026\)](#)
- ▶ **Watch:** [CNN: Waymo incidents raise safety concerns \(2026\)](#)
- ▶ **Watch:** [Global News: Meta and YouTube found liable \(2026\)](#)
- ▶ **Watch:** [CBS: what caused the CrowdStrike outage \(2024\)](#)
- ▶ **Watch:** [ITV News: what happened in the Post Office scandal \(2024\)](#)
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- ▶ **Watch:** [Kat Holmes: How inclusion shapes design](#)
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- ▶ **Watch:** [Hany Farid \(TEDx\): algorithms in the courtroom](#)
- ▶ **Watch:** [CNBC: AI and bias in criminal justice](#)
- ▶ **Watch:** [MIT 6.S191: algorithmic bias and fairness](#)
- ▶ **Watch:** [60 Minutes: Apple, the FBI, and the iPhone](#)

Reading

FROM THE ORIGINAL COURSE

- ↗ **Read:** [Suresh and Guttag: unintended consequences of ML](#)
- ↗ **Read:** [Mehrabi et al.: survey on bias and fairness in ML](#)
- ↗ **Read:** [Trewin: AI fairness for people with disabilities](#)
- ↗ **Read:** [IBM Research: AI Fairness 360](#)
- ↗ **Read:** [Howard and Borenstein: the ugly truth about our robot creations](#)
- ↗ **Read:** [The future of transportation: self-driving vehicles](#)
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- ↗ **Read:** [Apple accessibility design guidelines](#)
- ↗ **Read:** [Android: build more accessible apps](#)
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- ↗ **Read:** [The Wire: Aadhaar authentication errors](#)

Also: Obermeyer et al., Science 2019 · Kleinberg et al. 2016 · Chouldechova 2017 · Gebru et al., "Datasheets for Datasets" · Mitchell et al., "Model Cards" · Urbina et al., Nature Machine Intelligence 2022 · Kat Holmes, "Mismatch" (MIT Press, 2018) · Buolamwini and Gebru, "Gender Shades" (2018) · Raji and Buolamwini, "Actionable Auditing" (2019)

Every design is a decision.

Solve for one, extend to many.

Thanks

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